PTF ALL

EQUIP COOLING SUPPLY OFF Light Is On Switch At ALTN - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace Equipment Cooling Alternate Supply Fan [M1323]
Replace EQUIP COOLING SUPPLY Switch (Flight Deck) [S870]
Repair Wiring for D3472
Replace EQPT COOLING SUPPLY FAN CONTROL-ALTN Circuit Breaker [C1117]
Repair Wiring D3468-Circuit Breaker C1117
Replace Alternate Supply Fan Control Relay [R373]
Replace EQPT CLG SPLY FAN PWR-ALTN Circuit Breaker [C935]
Repair Wiring Between EQPT CLG SPLY FAN PWR-ALTN Circuit Breaker and Equipment Cooling Alternate Supply Fan
Cycle EQPT COOLING LOW FLOW DETECT SUPPLY Circuit Breaker
Clean Equipment Cooling Low Flow Sensor [M1915]
Replace Equipment Cooling Air Filter
Replace Equipment Cooling Low Flow Sensor [M1915]
Repair Wiring D3464-D11452
Repair Wiring D3464-L1141 Terminals
Replace Equipment Cooling Smoke Auto Reconfiguration Timer Relay [R1185]

A. Description

(1) The EQUIP COOLING SUPPLY OFF light on the equipment cooling panel is on with the EQUIP COOLING SUPPLY switch on ALTERNATE.

(2) The EQUIP COOLING SUPPLY OFF light comes on when the alternate supply fan does not blow enough cooling air through the equipment cooling duct. The monitoring circuit in the supply low flow detector then detects an inadequate air flow condition and provides a ground to allow the light to come on.

(3) This fault can also occur when power is first applied to the aircraft, the equipment cooling system is started, and these conditions occurred:

(a) The aircraft was parked in a very cold location, for many hours, with no power applied.

(b) The temperature in the aircraft was low.

1) If the OFF light came on for this condition, let the temperature increase in the Equipment Cooling Bay to more than 32F (0C). Then do the Fault Isolation Procedure below.

B. Initial Evaluation

(1) Open and then close this circuit breaker: C1433, EQPT COOLING LOW FLOW DETECT SUPPLY [P18-3, A17].

(a) Wait a minimum of ten minutes while you monitor the equipment cooling SUPPLY OFF light.

(b) If the light went off in ten minutes or less, then there was an intermittent fault.

(c) If the light stays on after ten minutes, then continue.

(2) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM and then set it to ALTN.

(3) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(4) If the light stays on after 20 seconds, then do these steps:

(a) Set the EQUIP COOLING SUPPLY switch to NORM.

(b) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(c) If the light goes off in less than 20 seconds, then do the Fault Isolation Procedure (Fan) below.

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(d) If the light stays on after 20 seconds then do the Fault Isolation Procedure (Airflow/Indication) below.

(5) If the light goes off in less than 20 seconds, then there was an intermittent fault.

C. Fault Isolation Procedure (Fan)

(1) Do this check of the circuits in the equipment cooling alternate supply fan (WDM 21-27-31):

(a) Disconnect connector D3472 from the equipment cooling alternate supply fan, M1323.

(b) For each of these pairs of pins of the connector on the equipment cooling alternate supply fan, do a check for the specified result:

1) Pin 5 to pin 6, the circuit is good if there is continuity.

2) Pin 2 to pin 3, the circuit is good if there is continuity.

3) Pin 1 to pin 3, the circuit is good if there is continuity.

4) Pin 1 to pin 2, the circuit is good if there is continuity.

(c) If the circuit was not good for one or more of the checks, then do these steps:

1) Replace the equipment cooling alternate supply fan, M1323. These are the tasks:

- Forward Equipment Cooling Supply Fan Removal, AMM TASK 21-27-02-000-801,
- Forward Equipment Cooling Supply Fan Installation, AMM TASK 21-27-02-400-801.

2) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.

3) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

4) Set the EQUIP COOLING SUPPLY switch to NORM.

5) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.

(d) If the circuit was good for all of the checks, then continue.

(e) Re-connect connector D3472 to the equipment cooling alternate supply fan, M1323.

(2) Do this check of the continuity through the EQUIP COOLING SUPPLY switch (flight deck) (WDM 21-27-31):

(a) Disconnect the wires from the EQUIP COOLING SUPPLY switch (flight deck), S870.

(b) For each switch position below, set the switch to the specified position and do a check for continuity between the specified pins on the switch:

1) Switch position ALTN, make sure there is continuity between wire terminal 5 and wire terminal 4.

(c) If there was a problem with the continuity, then do these steps:

1) Replace the EQUIP COOLING SUPPLY switch (flight deck), S870.

2) Re-connect the wires to the EQUIP COOLING SUPPLY switch (flight deck), S870.

3) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.

4) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

5) Set the EQUIP COOLING SUPPLY switch to NORM.

6) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.

(d) If there was not a problem with the continuity, then continue.

(e) Re-connect the wires to the EQUIP COOLING SUPPLY switch (flight deck), S870.

(3) Do this check of the voltage at the equipment cooling alternate supply fan:

(a) Disconnect connector D3472 from the equipment cooling alternate supply fan, M1323.

(b) Do a check for 28 VDC between pins 5 and 7 of connector D3472 (WDM 21-27-31).

(c) If 28 VDC is present, then do these steps:

1) Replace the equipment cooling alternate supply fan, M1323. These are the tasks:

- Forward Equipment Cooling Supply Fan Removal, AMM TASK 21-27-02-000-801,
- Forward Equipment Cooling Supply Fan Installation, AMM TASK 21-27-02-400-801.

2) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.

3) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

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4) Set the EQUIP COOLING SUPPLY switch to NORM.
5) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.

(d) If 28 VDC is not present, then continue.

(4) Do this check of the wiring:
(a) Do a wiring check to make sure that pin 7 of D3472 has continuity to ground (WDM 21-27-31).
(b) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D3472 to the equipment cooling alternate supply fan, M1323.
3) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.
4) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
5) Set the EQUIP COOLING SUPPLY switch to NORM.
6) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.

(c) If you do not find a problem with the wiring, then continue.
(d) Re-connect connector D3472 to the equipment cooling alternate supply fan, M1323.

(5) Do this check of the voltage at the alternate supply fan control relay:
(a) Disconnect connector D3468 from the alternate supply fan control relay, R373.
(b) Do a check for voltage between pin X1 of connector D3468 and ground (WDM 21-27-31). The correct voltage is 28 VDC.
(c) If the correct voltage is not present, then do this check of the voltage at the EQPT COOLING SUPPLY FAN CONTROL-ALTN circuit breaker:
1) Get access to the EQPT COOLING SUPPLY FAN CONTROL-ALTN circuit breaker, C1117 [P6-4, C13].
2) Do a check for voltage between terminal =L of the circuit breaker and ground (WDM 21-27-31). The correct voltage is 28 VDC.
3) If the correct voltage is not present, then do these steps:
a) Re-connect connector D3468 to the alternate supply fan control relay, R373.
b) Replace the EQPT COOLING SUPPLY FAN CONTROL-ALTN circuit breaker, C1117.
c) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.
d) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
e) Set the EQUIP COOLING SUPPLY switch to NORM.
f) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.
4) If the correct voltage is present, then do this check of the wiring:
a) Do a wiring check between this pin of connector D3468 at the alternate supply fan control relay and this terminal of circuit breaker C1117 (WDM 21-27-31):

D3468		**C1117**
pin X1	.....	term =L

b) If you find a problem with the wiring, then do these steps:
<1> Repair the wiring.
<2> Re-connect connector D3468 to the alternate supply fan control relay, R373.
<3> Close the EQPT COOLING SUPPLY FAN CONTROL-ALTN circuit breaker, C1117 [P6-4, C13].
<4> Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.
<5> Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
<6> Set the EQUIP COOLING SUPPLY switch to NORM.
<7> If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.

(d) If the correct voltage is present, then continue.

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(6) Replace the alternate supply fan control relay, R373.
   (a) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.
   (b) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
   (c) Set the EQUIP COOLING SUPPLY switch to NORM.
   (d) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.
   (e) If the light stayed on after 20 seconds with the switch at ALTN, then continue.

(7) Do this check of the voltage supplied for the EQPT CLG SPLY FAN PWR-ALTN circuit breaker:
   (a) Get access to the EQPT CLG SPLY FAN PWR-ALTN circuit breaker, C935 [P91, A8].
   (b) Do a check for three-phase 115 VAC between terminals A2, B2, and C2 of the circuit breaker and ground (WDM 21-27-31).
   (c) If three-phase 115 VAC is not present, then do these steps:
      1) Replace the EQPT CLG SPLY FAN PWR-ALTN circuit breaker.
      2) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.
      3) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
      4) Set the EQUIP COOLING SUPPLY switch to NORM.
      5) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.
   (d) If three-phase 115 VAC is present, then continue.
   (e) Close any access that you made to get to the EQPT CLG SPLY FAN PWR-ALTN circuit breaker, C935 [P91, A8].

(8) Do a check of the wiring between the EQPT CLG SPLY FAN PWR-ALTN circuit breaker, C935 and the equipment cooling alternate supply fan, M1323 (WDM 21-27-31).
   (a) If you find a problem with the wiring, then do these steps:
      1) Repair the wiring.
      2) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN.
      3) Wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
      4) Set the EQUIP COOLING SUPPLY switch to NORM.
      5) If the light was off, or if it went off in less than 20 seconds with the switch at ALTN, then you corrected the fault.

D. Fault Isolation Procedure (Airflow/Indication)

(1) Open and then close this circuit breaker: C1433, EQPT COOLING LOW FLOW DETECT SUPPLY [P18-3, A17].
   (a) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
   (b) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
   (c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.
   (d) If the light stayed on after 20 seconds, then continue.

(2) Clean the equipment cooling low flow sensor. This is the task:
   • Equipment Cooling Low Flow Sensors Cleaning, AMM TASK 21-27-03-400-802.
   (a) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
   (b) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
   (c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.
   (d) If the light stayed on after 20 seconds, then continue.

(3) Replace the equipment cooling air filter. These are the tasks:

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- Equipment Cooling Air Filter Removal, AMM TASK 21-27-01-000-801,
- Equipment Cooling Air Filter Installation, AMM TASK 21-27-01-400-801.

(a) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(b) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If the light stayed on after 20 seconds, then continue.

(4) Replace the equipment cooling low flow sensor, M1915. These are the tasks:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-801,
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-801.

(a) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(b) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If the light stayed on after 20 seconds, then continue.

(5) Do this check of the wiring:

(a) Disconnect connector D3464 from the equipment cooling low flow sensor, M1915.

(b) Disconnect connector D11452 from the power distribution panel 2, P92.

(c) Do a wiring check between this pin of connector D3464 at the equipment cooling low flow sensor and this pin of connector D11452 at the power distribution panel 2 (WDM 21-27-31):

D3464		**D11452**
pin 9	.....	pin 7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D3464 to the equipment cooling low flow sensor, M1915.
3) Re-connect connector D11452 to the power distribution panel 2, P92.
4) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
5) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
6) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Re-connect connector D11452 to the power distribution panel 2, P92.

(6) Do this check of the wiring:

(a) Disconnect the wires from the equipment cooling SUPPLY OFF light, L1141.

(b) Do a wiring check between this pin of connector D3464 at the equipment cooling low flow sensor and these wire terminals at the equipment cooling SUPPLY OFF light (WDM 21-27-31):

D3464		**L1141**
pin 9	.....	term 2

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D3464 to the equipment cooling low flow sensor, M1915.
3) Re-connect the wires to the equipment cooling SUPPLY OFF light, L1141.
4) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

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5) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.
6) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D3464 to the equipment cooling low flow sensor, M1915.

(f) Re-connect the wires to the equipment cooling SUPPLY OFF light, L1141.

(7) Replace the equipment cooling smoke auto reconfiguration timer relay, R1185.

(a) Set the EQUIP COOLING SUPPLY switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(b) Set the EQUIP COOLING SUPPLY switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling SUPPLY OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

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